

Making PDFs Accessible with AI

Peya Mowar
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Introduction to Accessibility and Assistive Technology
CMU 05332/05632/17332/17732 Fall 2026

iTagPDF: Towards Finally Automating PDF Accessibility

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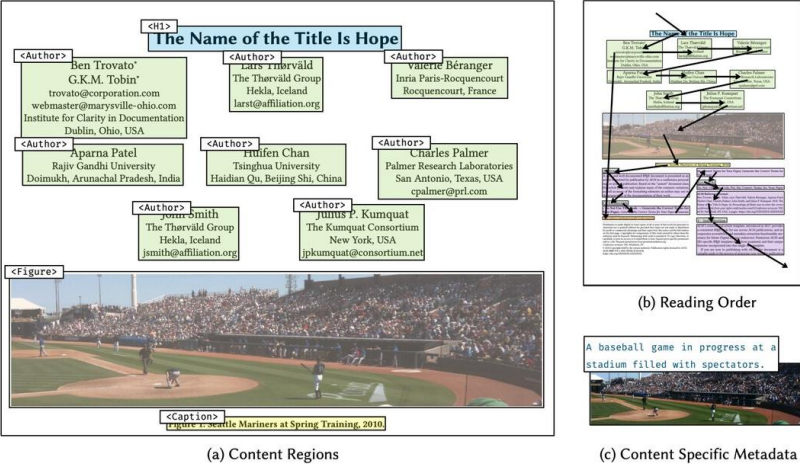


Figure 1: Intelligent PDF Tagging (iTagPDF) automatically produces three kinds of PDF accessibility metadata: (a) content regions and their locations (tags), (b) their reading order, and (c) content specific metadata, such as structure information for tables and alt texts for figures and formulas. iTagPDF is the first to combine all major tasks of making a PDF accessible into one tool, and outperforms all prior approaches by jointly modeling the input source and the rendered pixels of the PDF.

Abstract

Most academic research is ultimately disseminated through documents in the PDF format. This format has advantages in flexibility and portability, but presents challenges for accessibility that have stubbornly resisted solutions despite decades of attempts. Tagging PDFs is hard to automate because tags are currently generated visually, not semantically, which makes

the output cluttered and manual correction tedious and error-prone. Ironically, this semantic structure already exists during authoring but is discarded during PDF rendering. This raises an obvious question, can we use this lost semantic information to better automate tagging in PDFs? In this paper, we develop iTagPDF, a system that refines generated metadata using the semantics in the source documents of research papers. We demonstrate that the metadata generated by iTagPDF already surpasses what authors currently submit to ACM conferences on many criteria. Our approach represents a concrete step toward finally automating accessibility remediation in research paper PDFs.



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<https://doi.org/10.1145/3772318.3790289>

Today's goals

By the end of class, you will be able to:

- Explain what PDF accessibility entails and its challenges
- Design an approach for automating PDF accessibility
- Inspect automated tagging and identify semantic errors
- Describe how iTagPDF improves automated tagging



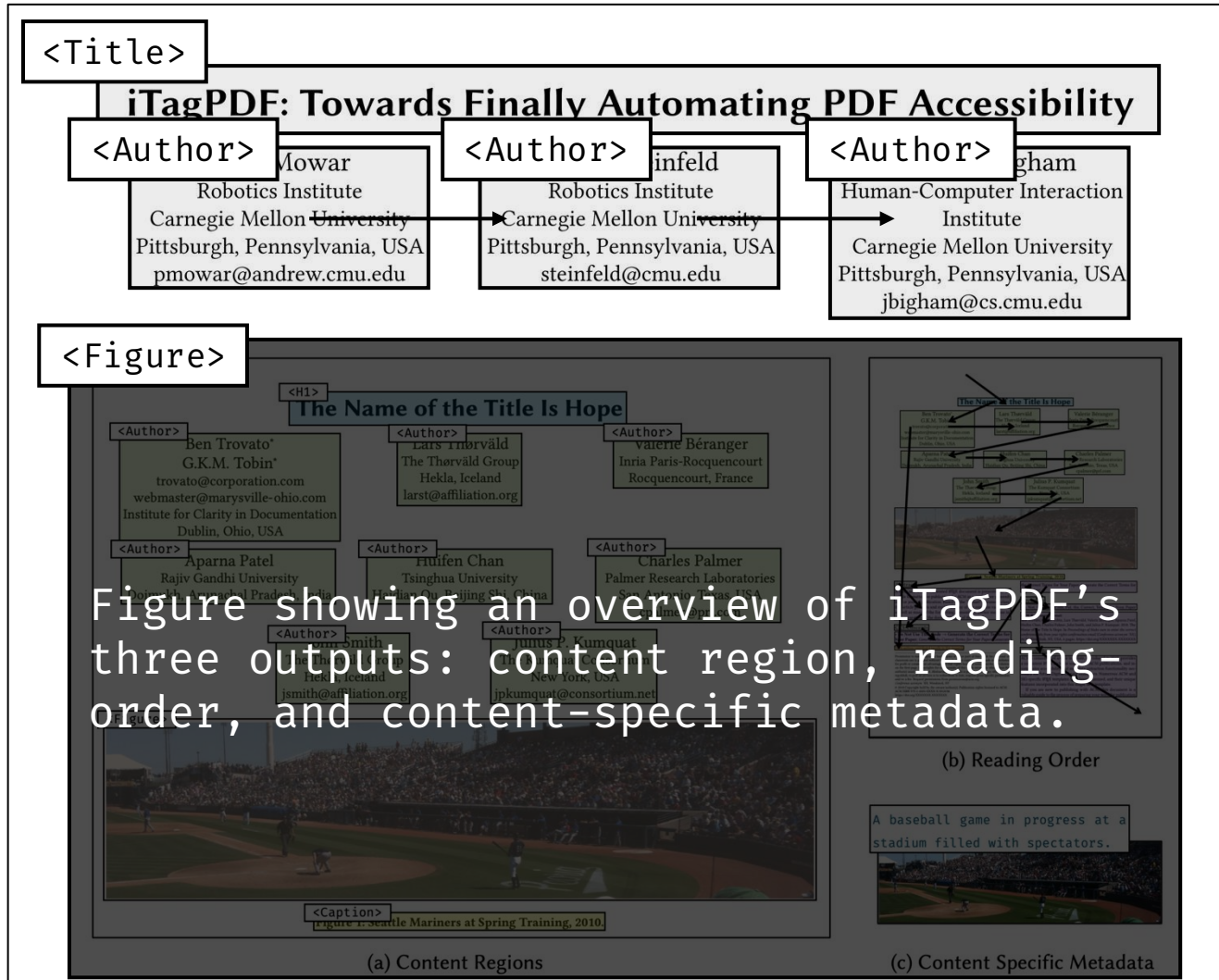
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AI + Accessibility

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How do we make a PDF accessible?



By providing accessibility metadata:

- Content regions and tags
- Reading order
- Content-specific metadata (e.g., alt-texts)

What does accessibility metadata help with?

iTagPDF: Towards Finally Automating PDF Accessibility

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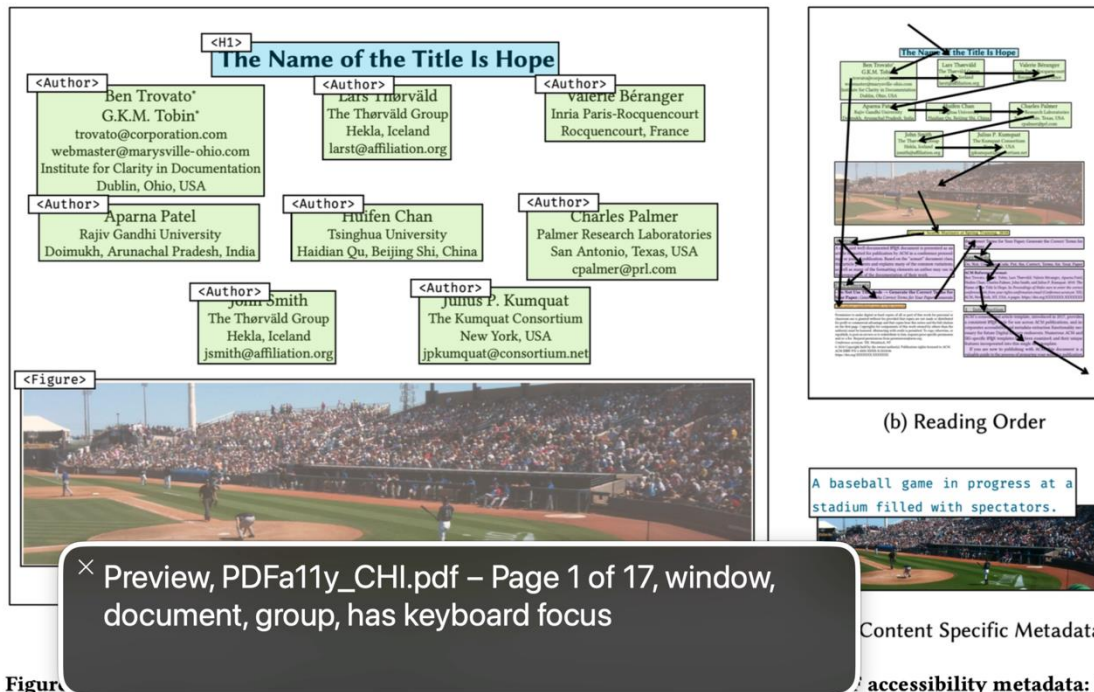


Figure 1: iTagPDF accessibility metadata: (a) content regions and their locations (tags), (b) their reading order, and (c) content specific metadata, such as structure information for tables and alt texts for figures and formulas. iTagPDF is the first to combine all major tasks of making

Without this accessibility metadata:

- No tags or labels for text
- Incorrect reading order
- Figures may be skipped

PDF remediation tools help provide this metadata.

itagpdf-chi26.pdf

Create

Add file to PDF SpaceAll toolsEditConvertE-Sign

Find text, tools, or help

ShareAsk AI Assistant

H1

iTagPDF: Towards Finally Automating PDF Accessibility

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<H1>

The Name of the Title Is Hope

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<Figure>



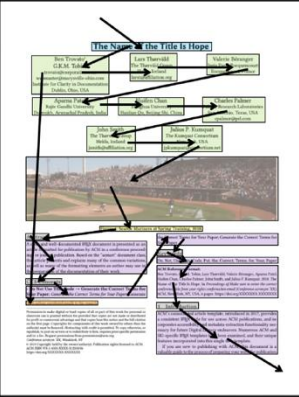
<Caption>

figure 1: seattle mariners at spring training, 2010.

(a) Content Regions

The Name of the Title Is Hope

New Yorker G.K.M. Tobin's...
Lars Thorsvald's...
variee Béranger's...
Aparna Patel's...
Huiwen Chan's...
Charles Palmer's...
John Smith's...
Jumus P. Kumquat's...



(b) Reading Order

A baseball game in progress at a stadium filled with spectators.



(c) Content Specific Metadata

Accessibility tags

Tags<Document>

Reading Order

Draw a rectangle around the content then click one of the buttons below:

Text/Paragraph

Form Field

Heading 1

Heading 2

Heading 3

Reference

Figure

Figure/Caption

Table

Cell

Formula

Note

Background/Artifact

Table Editor

Show page content groups

Page content order

Structure types

Show table cells

Display like elements in a single block

Show tables and figures

Clear Page StructureShow Order Panel

Adobe Acrobat Pro

But they lack quality automation.

[illegible]



Homework

Homework 1: Accessible Documents

Parts

[Homework 1A: Create an Accessible PDF](#)

[Homework 1B: Create an Accessible Presentation](#)

Homework 2: Disability Deep Dive

Homework 3: Try Out Assistive Technology

Homework 4: Accessibility Evaluation

Homework 1A: Create an Accessible PDF

Deliverables

Accessible PDF - 50 points

Getting Started

Have you faced any challenges in making PDFs accessible?

If you've ever used Adobe Acrobat, you know that if you don't have a Creative Cloud license, you have two options to access Acrobat.

1. Use [Virtual Andrew](#).
 - a. Log into Virtual Andrew from your [web browser](#) or through the [Omnissa Horizon client application](#).
 - b. Once you have logged into the Virtual Windows account, find Acrobat in the Start Menu and run it.
2. Install Adobe Creative Cloud with a license that we've given you for this class.
 - a. Go to the [Creative Cloud website](#) and click Sign In.
 - b. Enter your Andrew email address when prompted.
 - c. Click Continue and then, if prompted, click Company or School Account.
 - d. Enter your Andrew userID and password at the CMU Web Login screen.
 - e. **If you have trouble logging in, please [submit a help ticket to CMU IT](#).**
 - f. Click Install Creative Cloud App.
 - g. Double-click the installer file and follow the instructions.

On This Page

[Deliverables](#)

[Getting Started](#)

[Goal](#)

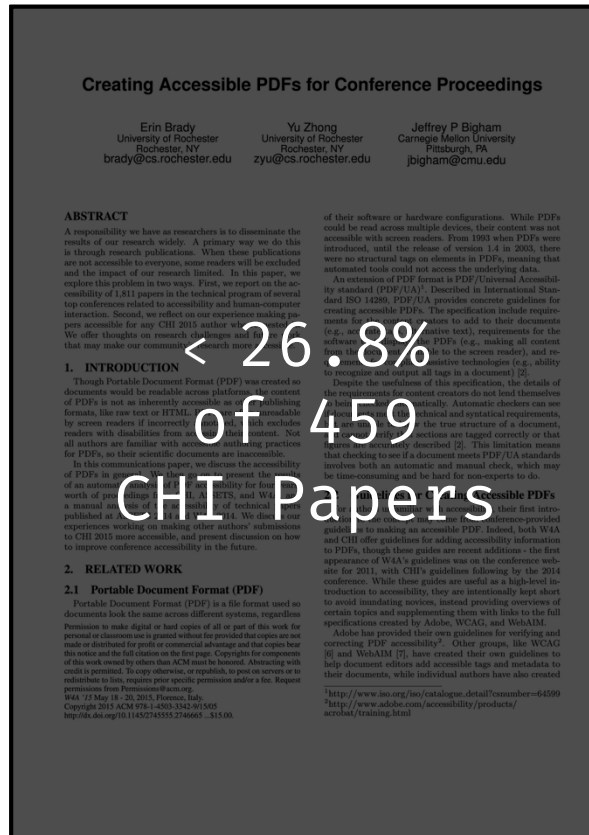
[Instructions](#)

[Check for Accessibility](#)

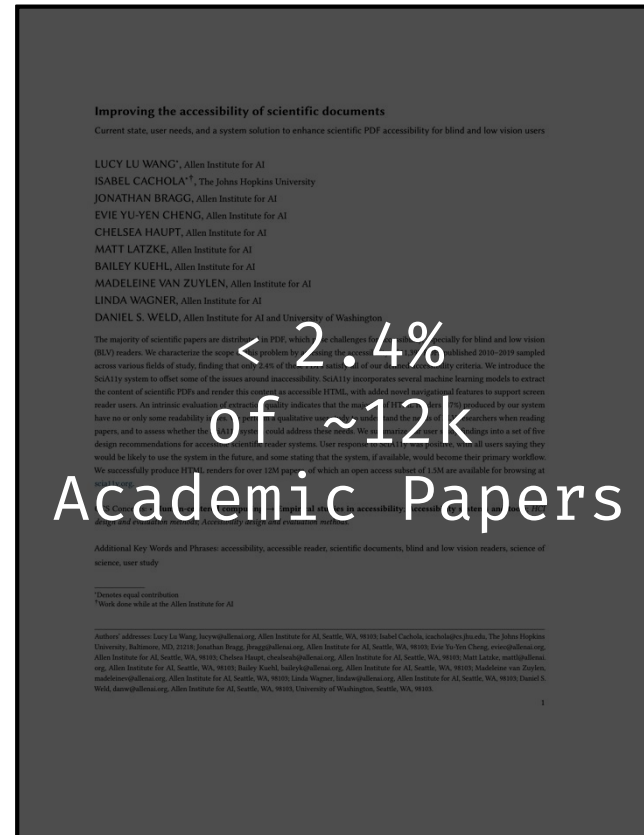
[Fix Accessibility Issues](#)

[Submission](#)

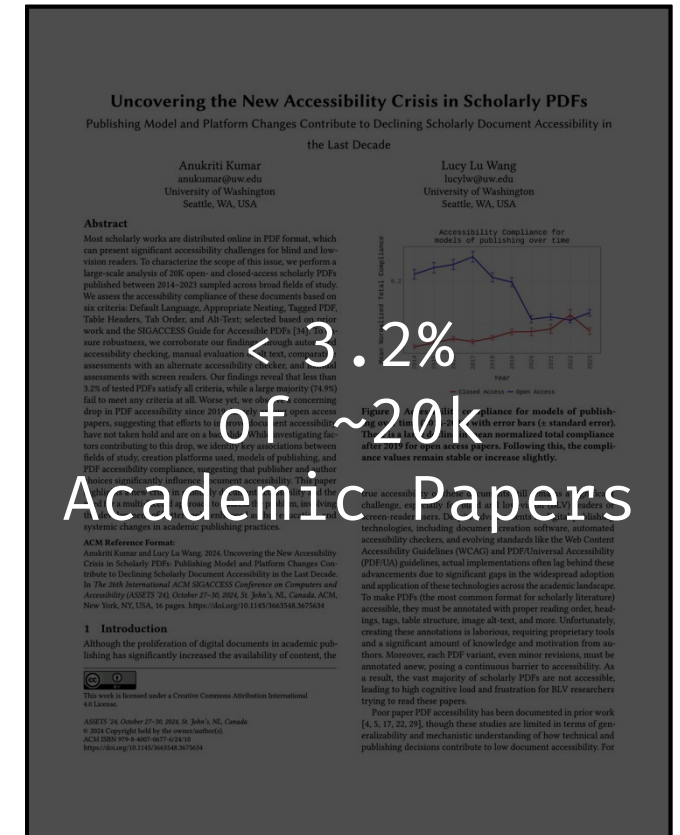
Remediating PDFs is a pain, so we don't.



Brady et al., 2015



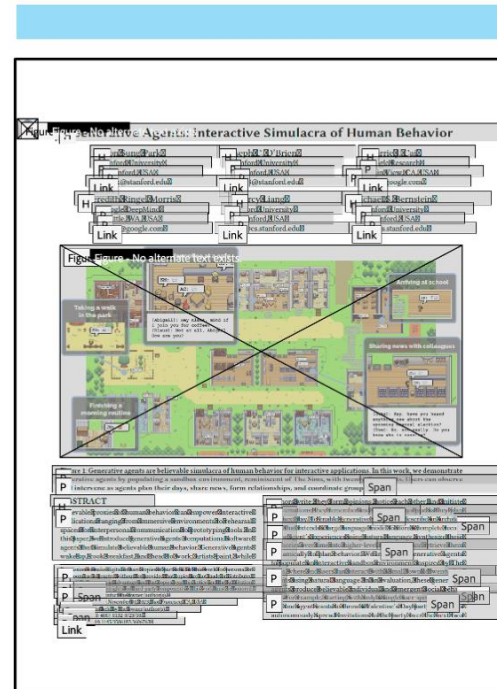
Wang et al., 2021



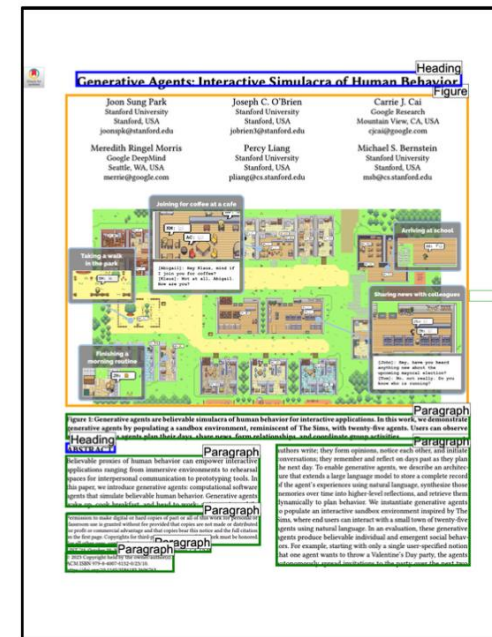
Kumar et al., 2024

Approaches I tried

Accessibility Remediation Tools



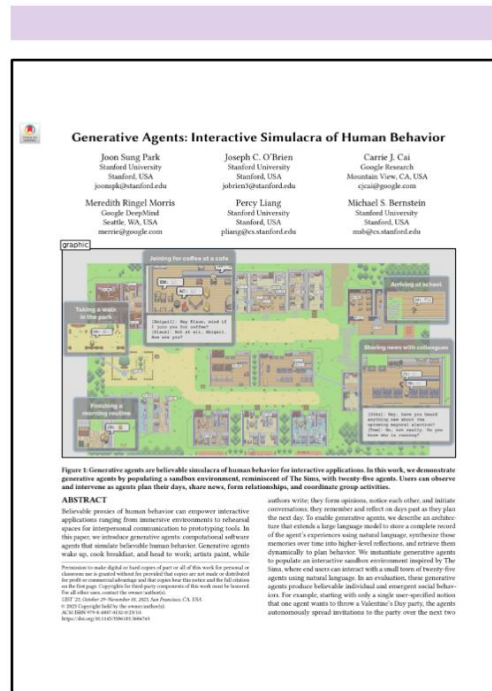
(i) Acrobat
Auto-Tagging



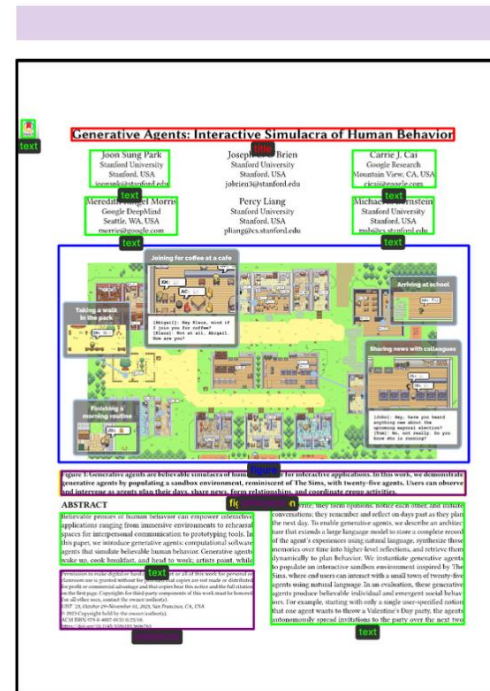
(v) PAVE v2

Approaches I tried

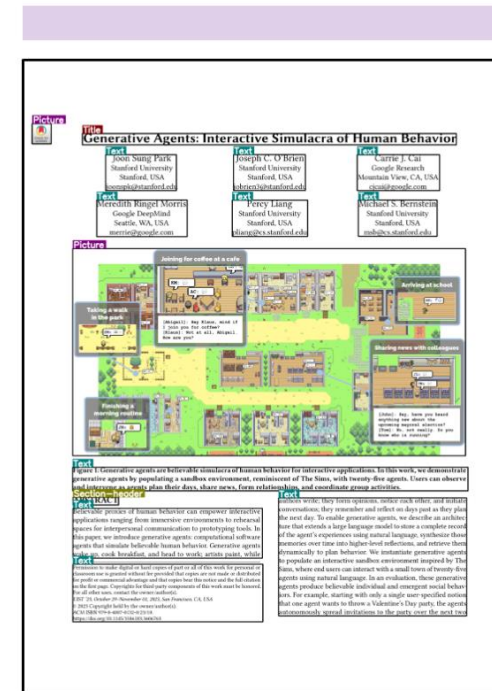
Machine Learning Pipelines for Document Understanding



(ii) GROBID
(used in SciA11y)



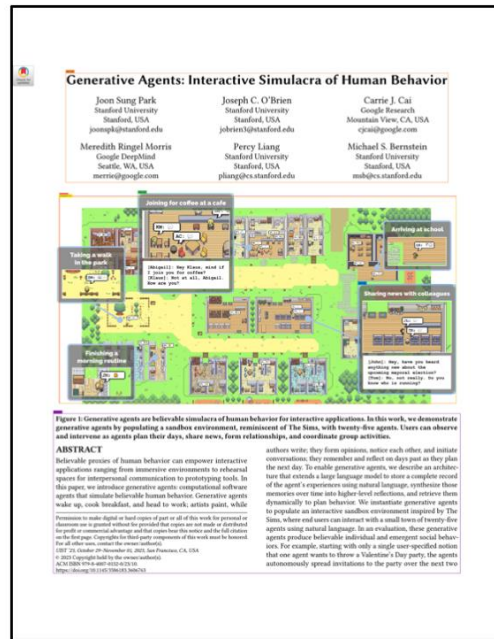
(iii) Paddle OCR



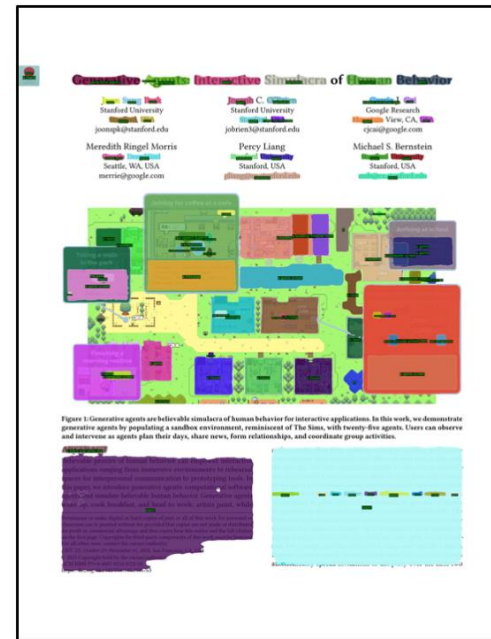
(iv) YOLO-v11
(DocLayNet)

Approaches I tried

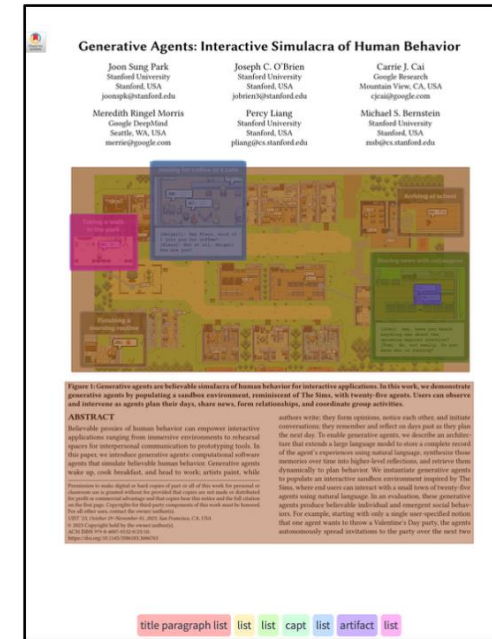
Vision Language / Multimodal Models



(vi) YOLO-World



(vii) Semantic-SAM



(viii) Grounding DINO

L^AT_EX

```
\title{iTagPDF}{iTagPDF: Towards Finally Automating  
PDF Accessibility}
```

```
\author{Peya Mowar}  
\affiliation{%  
\institution{Carnegie Mellon University}  
\city{Pittsburgh}  
\state{Pennsylvania}  
\country{USA}}  
\email{pmowar@andrew.cmu.edu}
```

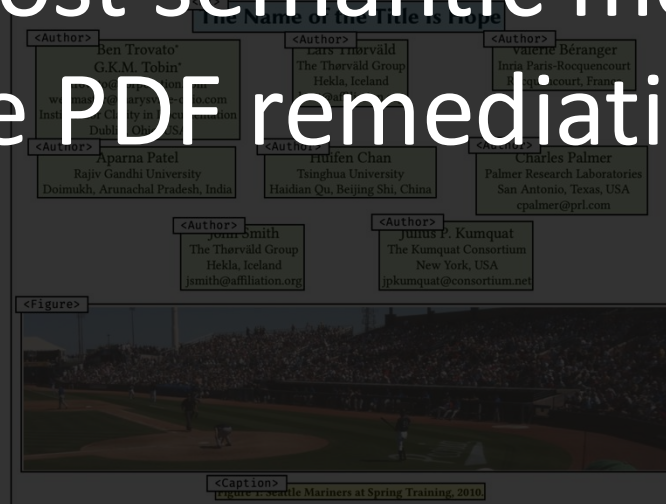
Can we preserve this lost semantic metadata
to better automate PDF remediation?

iTagPDF: Towards Finally Automating PDF Accessibility

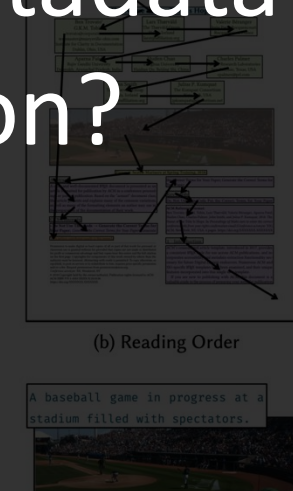
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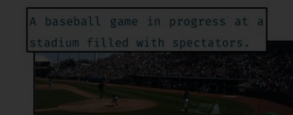
Jeffrey P Bigham
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jbigham@cs.cmu.edu



(a) Content Regions

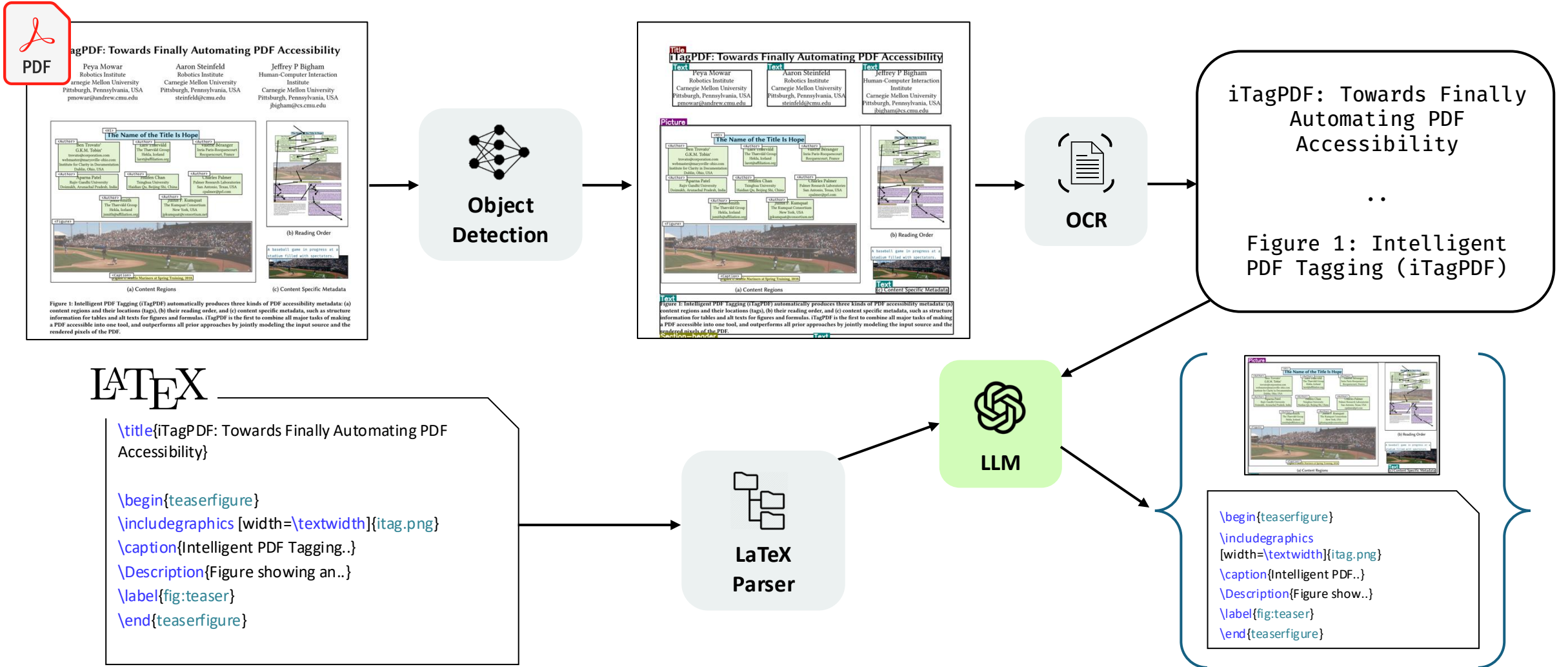


(b) Reading Order

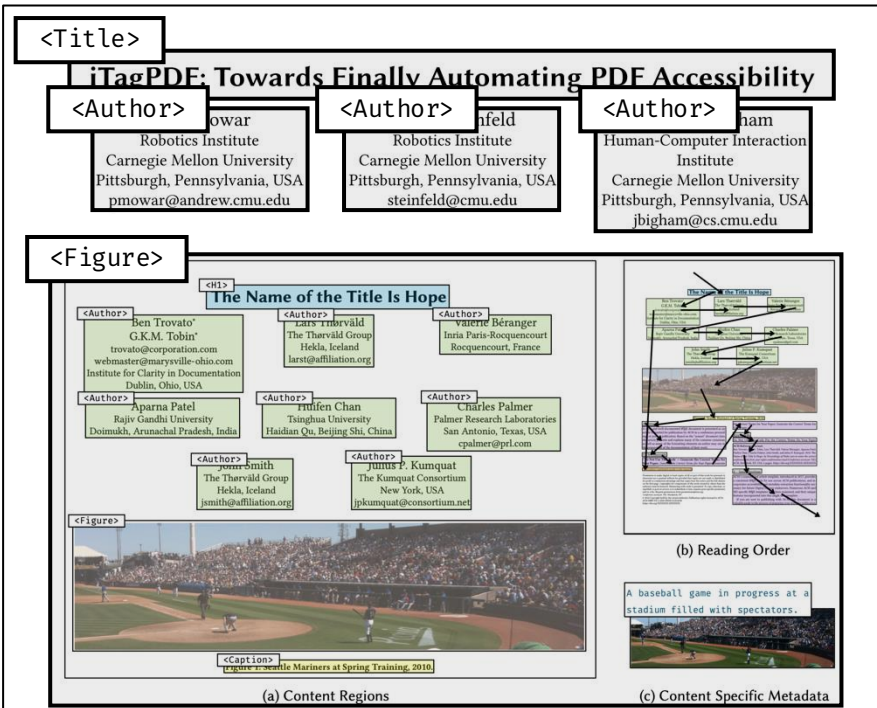


(c) Content Specific Metadata

Preserving Accessibility Metadata



Preserving Accessibility Metadata

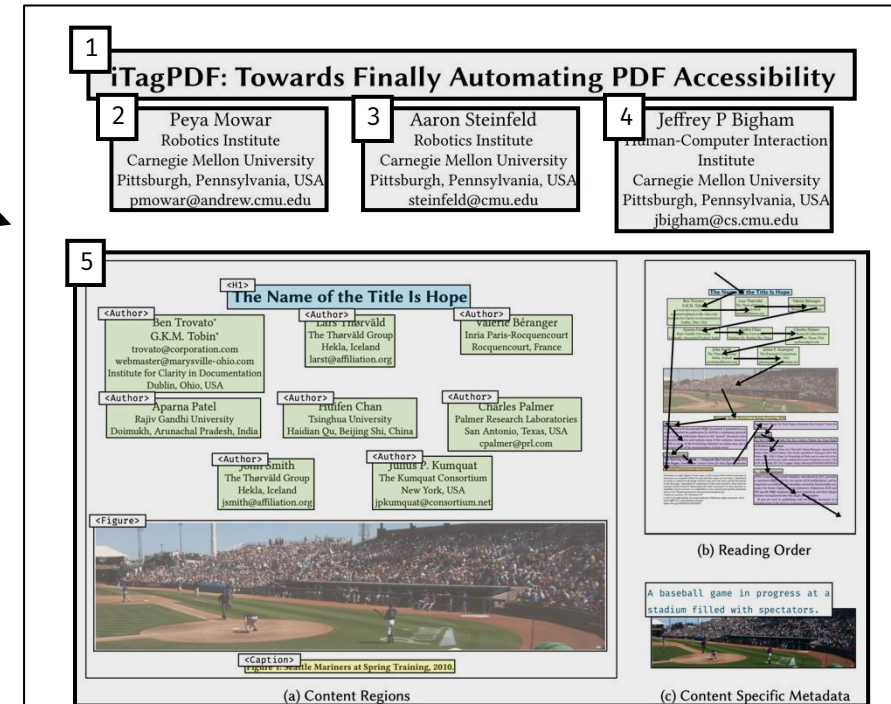


Content Regions and Tags

<Figure>

Figure showing an overview of iTagPDF's three outputs: content region, reading-order, and content-specific metadata.

Content-specific metadata



Reading Order

```
\begin{teaserfigure}
\includegraphics
[width=\textwidth]{itag.png}
\caption{Intelligent PDF..}
\Description{Figure show..}
\label{fig:teaser}
\end{teaserfigure}
```

iTagPDF - Demo



The screenshot shows the iTagPDF web interface. At the top left, there is a logo with a document icon and the text "iTagPDF". Below it, it says "Carnegie Mellon University · Project". At the top right, there is a circular profile icon with the letter "P". The main content area has a heading "Make your paper accessible" followed by a paragraph: "Automatically produces PDF accessibility metadata — content regions and their tags, reading order, and content-specific metadata such as alt texts for figures." Below this, the text "try it yourself:" is displayed in a large, white, sans-serif font. Underneath, there is a dashed rectangular box containing an upload icon (an arrow pointing up) and the text "Drop ZIP or click to browse" and "PDF or Text source for file". Overlaid on this box is the URL "www.itagpdf.com" in a large, white, sans-serif font. Below the dashed box, there are two links: "Using your saved API key." and "Use a different key.". At the bottom, there is a button labeled "Run Pipeline".

iTagPDF
Carnegie Mellon University · Project

Make your paper accessible
Automatically produces PDF accessibility metadata — content regions and their tags, reading order, and content-specific metadata such as alt texts for figures.

try it yourself:

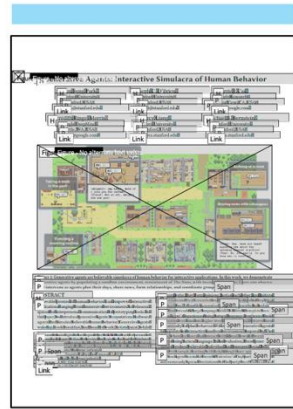
Drop ZIP or click to browse
PDF or Text source for file

www.itagpdf.com

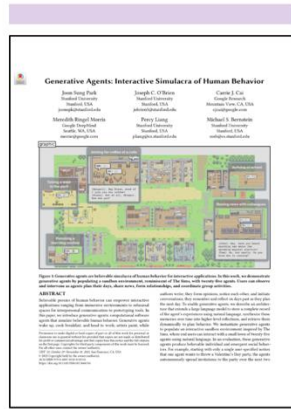
Using your saved API key. [Use a different key.](#)

Run Pipeline

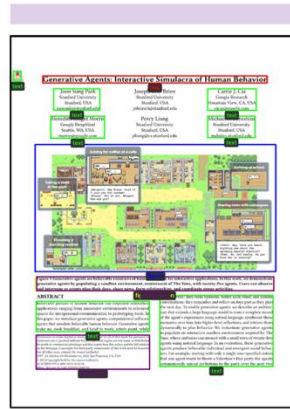
iTagPDF gave good and consistent results!



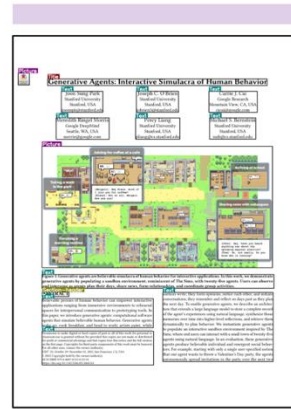
(i) Acrobat Auto-Tagging



(ii) GROBID (used in SciA11y)



(iii) Paddle OCR



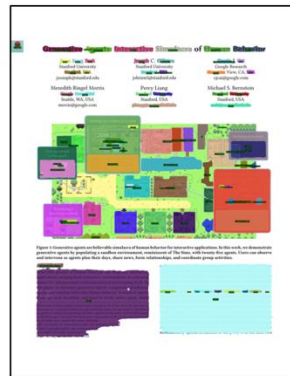
(iv) YOLO-v11 (DocLayNet)



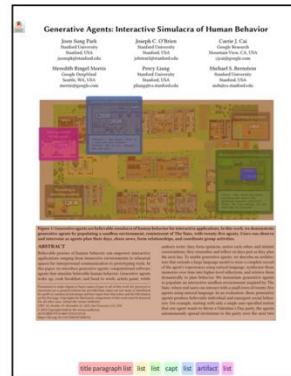
(v) PAVE v2



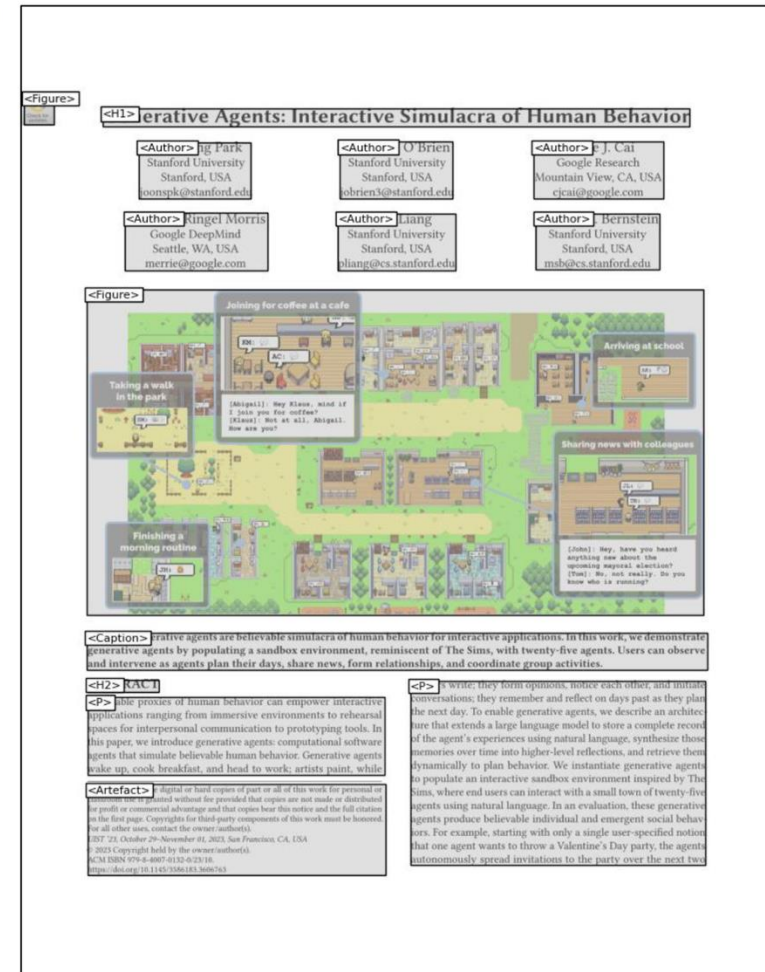
(vi) YOLO-World



(vii) Semantic-SAM



(viii) Grounding DINO



(ix) iTagPDF (Ours)

Natively Tagging PDFs in the TeX Environment

The screenshot shows the Overleaf website interface. At the top is a dark navigation bar with the Overleaf logo on the left and links for Product, Solutions, Templates, Pricing, Help & resources, Projects, and Account on the right. Below the navigation bar is a light gray sidebar on the left containing a search bar and a list of links organized into categories: Documentation Home, Overleaf Learning Center, Overleaf guides, LaTeX basics, and Mathematics. The main content area on the right displays an article titled 'An introduction to tagged PDF files: internals and the challenges of accessibility'. The article includes an update date of January 2026, a link to instructions on creating tagged PDFs, and a summary of new features for automatic PDF tagging in TeX Live 2025. A 'Contents' section at the bottom lists four items: 1. Update: January 2026, 2. What does this article cover?, 3. Introduction, and 4. Genesis of PDF as final-form digital paper.

Overleaf

Product Solutions Templates Pricing Help & resources Projects Account

Search help library...

An introduction to tagged PDF files: internals and the challenges of accessibility

Update: January 2026

Please see [Overleaf's instructions on creating tagged PDFs](#).

The [LaTeX](#) team has released new features that enable automatic PDF tagging, the central requirement for PDF accessibility. These new features are available in Overleaf in [TeX Live 2025](#), with more recent updates available in the rolling TeX Live release available through [Overleaf Labs](#).

It is now possible to produce a tagged PDF from LaTeX source that is conformant with WCAG 2.1 Level AA by following our [user docs](#) using TeX Live releases available in Overleaf.

The article below from 2020, while not updated with the latest information on LaTeX accessibility, can continue to be read for interest and for background information on PDF tagging.

Contents

1. Update: January 2026
2. What does this article cover?
3. Introduction
4. Genesis of PDF as final-form digital paper

- LaTeX Tagged Project:
started 6 years ago
- New updates in 2026
- Currently works for limited
packages and compilers

Claude's Cowork has now started asking about PDF source for tagging!

ACM_Confer...
late__5_.pdf

PDF

can you tag this pdf for accessibility

Ran 4 commands · 1 note

Ran a command

Ran a command

Ran a command

Asking a question...

15s

This is the 6-page ACM acmart template from pdfTeX — currently untagged (no StructTreeRoot, no /MarkInfo). Do you have the LaTeX source?

< 1 of 3 > ×

1 No, PDF only

I'll retrofit tags onto the existing PDF: insert marked content into the page streams and build a structure tree. →
Heuristic but works on any PDF.

2 Yes, I can share the .tex

Best fidelity: recompile with LaTeX's \DocumentMetadata tagging (phase-III / PDF/UA-2). Real tags from the source semantics, not guesses.

3 Do the PDF retrofit now, source later

Retrofit this copy so you can see the result, and I'll note what the source route would fix.

Something else

Skip

Or reply directly...

+ 🗨 Manual ▾

Opus 5 High 🔊 □

⬆ ⬇ to navigate · ⬅ to select · or type below

OpenAI's GPT-6 Astra can now tag PDFs?

 **ACM_Conference_Proceedings_Pri...**
PDF

can you tag this PDF for accessibility

Worked for 7m 28s ▾

I'll inspect the PDF, then add accessibility tags and check its reading order. I'm using the PDF and Library skills to prepare the finished file.

🔍 Read files, ran commands

The six-page PDF has no accessibility tags. It includes two-column text, two figures, two tables, equations, and links. I'll tag those elements, add figure descriptions, and set the reading order while preserving the page appearance.

📄 Ran commands

The tags are in place, including table headers, figure descriptions, equation descriptions, and all 96 link annotations. All six pages render identically to the original. I'm now checking the tag structure and reading order for misplaced or empty elements.

📖 Used Library, ran commands ▾

📄 Inspected, improved, validated, and reviewed PDF accessibility structure

📖 Replaced PDF with accessibility tags and structure

📄 Applied PDF file metadata using a library

Done—added accessibility tags, reading order, figure descriptions, table headers, and tagged links. The original layout is preserved. Full PDF/UA compliance  not been independently certified.

 **ACM_Conference_Proceedings_Accessibility_Tagged.pdf**

 **ACM_Conference_Proceedings_Accessibility_Tagged...**
PDF

You're out of Work usage for now
Upgrade or add credits to keep using Work now, or wait for your usage to reset at 11:52 PM

UpgradeAdd credits

+ Work on anything

GPT-6 Astra Light ▾

A wide-angle shot of a baseball game in progress. The field is visible with players in position. The stands are filled with a large crowd of spectators. The sky is clear and blue. The text "can now tag accurately" is overlaid on the right side of the image in a large, white, sans-serif font.

can now tag accurately

How does OpenAI's GPT-6 Astra tag PDFs?

- Predominantly reliant on tool use:
 - PyMuPDF for understanding PDF layout and structure
 - PDFPlumber and PyPDF for embedding tags in the PDF
- Some underlying semantic understanding
 - Assigns labels and reading order based on its own (black box)
 - Can fumble in these aspects
 - iTagPDF is more reliable here because it literally cheats from the source!

Thank you!
Any Questions?